SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

(Deemed to be University)

Department of Mathematics and Computer Science

SYLLABUS - UMAT-604 (TPDE) - Theory of Partial Differential Equations

L – T – P – C	4 – 0 – 0 – 4	**Prerequisites**	—	Total Periods: **56**
Course Category	DSE			

Course Objectives

Using the tools of calculus, the course covers the basic methods of solving linear and non-linear partial differential equations. No assumptions on the knowledge of Analysis is made. Fundamental equations such as Laplace, Heat, Transport and Wave equations are discussed in detail. Fourier Transform in the higher dimensions is also introduced. For solving first order partial differential equations with real coefficients, the method of characteristics is employed.

Course Outcomes

- Students will be able to classify a given partial differential equation.
- Students can discuss the role of fundamental solution and its utility for basic equations such as Laplace, Heat and Wave equations.
- Students learn an approach to solve partial differential equations through transform techniques. In particular, the student learns a higher dimensional Fourier Transform.
- Students learn the method of characteristic and its geometric insights to solve a first order partial differential equation with real coefficients.
- Students would be able to appreciate the role of symmetry in solving partial differential equations.

Syllabus

Unit	Topic	Contents	Periods
0	**Review of Calculus Facts**	Boundaries, Gauss-Green theorem, Polar coordinates, Coarea formula	3
I	**Convolution and Smoothing**	Convolution and smoothing	4
II	**Single and Systems of PDEs**	Singles and Systems of PDEs and Strategies for studying PDE	3
III	**Transport Equations**	IVP, Nonhomogeneous	3
IV	**Laplace's Equation**	Fundamental solution, mean-value formulas, properties of harmonic functions, Green's function, energy methods	10
V	**Heat Equation**	Fundamental solution, mean-value formulas, properties of solution, energy method	9

Unit	Topic	Contents	Periods
VI	**Wave Equations**	Solution of wave equation, d'Alembert's formula and Spherical means	4
VII	**Non-linear First-order PDE**	Complete integrals, envelops, characteristics	10
VIII	**Similarity Methods and Transform Methods**	Fourier transform, Laplace transform, Converting nonlinear into linear PDE	10
		Total	**56**

Key Text(s)

- Lawrence C. Evans, Partial Differential Equations, Graduate Studies in Mathematics, Volume 19, 1997. Ch: 1, 2 (sec 2.1 to 2.3 and 2.4.1), 3 (sec 3.1 to 3.2), 4 (sec 4.2, 4.3, 4.4), Appendix C.1, C.2, C.3, C.4, D.5 (theorem, no proof), Chapter 11 (theorem 1, no proof), 14 and 17.

Suggested/Additional Readings

- Francois Treves, Basic Linear Partial Differential Equations, Dover Publications, Inc.